

Glitchwave 567 — CONTROL board

Assembly instructions for PCBWay

Customer: Illicit Apothecary
Project: Glitchwave 567 guitar pedal — CONTROL board
Board file base name: glitchwave567_ctrl
Revision date of this package: 2026-08-02
Quantity: 5 assembled boards, full turnkey

This is the second of two boards. The other is `glitchwave567` (MAIN). They stack together.

1. Files in this submission

File	What it is
glitchwave567_ctrl_gerbers.zip	Gerbers (RS-274X) + Excellon drill, PTH and NPTH separate, plus drill maps
BOM_ctrl_pcbway.csv	Bill of materials, 11 lines
CENTROID_ctrl_pcbway.csv	Pick-and-place, 22 placements — 21 top, 1 bottom
this PDF	Assembly notes

2. Board specification

Identical stackup to the MAIN board.

Parameter	Value
Size	138 x 114 mm
Layers	4
Stackup, top to bottom	L1 F_Cu signal / L2 In1_Cu solid ground plane / L3 In2_Cu signal / L4 B_Cu signal
Thickness	1.6 mm
Copper	1 oz all layers
Minimum track / clearance used	0.25 mm / 0.13 mm — please build to 5/5 mil
Vias	0.5 mm pad, 0.3 mm drill. No via-in-pad on this board.
Surface finish	HASL lead free
Mounting holes	four Ø3.2 mm NPTH at (17, 9.5), (121, 9.5), (17, 108.5), (121, 108.5)

3. Things that will otherwise generate a query

3.1 This board is assembled on BOTH sides

Potentiometers, foot switches and the WS2812 LEDs are on the **top** (`F_Cu`).

J1, the 2x8 board-to-board header, is on the BOTTOM (`B_Cu`). This is deliberate — it faces down into the gap and plugs into the socket on the MAIN board below. If J1 is fitted to the top side the two boards cannot mate and both boards are scrap.

J1 uses a project-specific pre-mirrored footprint, `PinHeader_2x08_P2.54mm_Vertical_Mirrored` , so that pin N lands on pin N of the MAIN board's socket. Please place strictly per the centroid file and do not re-derive the pin order.

3.2 Do-not-populate — foot switches

SW1 and SW2 must not be fitted and must not be sourced. They are 3PDT panel-mount foot switches, supplied by the customer and fitted at final assembly. They are marked DNP in the BOM. Leave the pads bare.

3.3 No substitutions

Every populated BOM line carries a specific LCSC part number. Please do not substitute without written approval. The potentiometers in particular (ALPS RK09K1130A70, 30 mm shaft) are dimensionally critical to the enclosure — a same-value pot with a different shaft length will not fit.

4. Assembly summary

- 5 distinct part numbers populated, 12 SMD placements, 8 through-hole placements.
 - No BGA, QFN or QFP parts. No X-ray inspection required.
 - Boards ship as single pieces, not panelised.
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5. Contact

Questions on this package should go back to the customer before production starts, not after.